Lesson 23 Pigeonhole Principle

In Lessons 23 to 26, the last part of the course, we bring together sets, functions and counting. In this lesson, we explore the **Pigeonhole principle**.

The Pigeonhole Principle (PHP)
Let A and B be finite sets with $|A| = n$ and $|B| = m$. If $n > m$, then there are no one-to-one functions from A to B because there are not enough different elements in B to be assigned to every element in A.

Think of A as a set of pigeons and B as a set of pigeonholes. The Pigeonhole Principle states that if there are more pigeons than holes, then some pigeons must share holes. This seemingly trivial statement is surprisingly powerful. We give a few examples.

In a class of 40 students, there are two students who were born in the same month.

Proof Let the 12 birth months be the pigeonholes and the 40 students be the pigeons. There are more pigeons than holes, so there is a hole with more than one pigeon, i.e. there must be more than one student born in the same month.

Let $n \in \mathbb{N}$. There are distinct positive integers a and b such that $n^a - n^b$ is divisible by 10.

Proof Let the ten possible ones digits be the holes and the eleven numbers $n^1, n^2, n^3, \ldots, n^{11}$ be the pigeons. Two of the numbers, say n^a and n^b, must have the same ones digit. Then $n^a - n^b$ has 0 as the ones digit and is divisible by 10.

Show that two people in Boston have the same number of hair.

Proof The number of people in Boston is greater than the number of hair a person may have.

Show that when we select 5 points in a 2×2 square, there are always two points whose distance is at most $\sqrt{2}$.

Proof Divide the 2×2 square into four 1×1 squares. Since there are 5 points but only 4 squares, there are at least two points in one of the squares, and the distance between these two points are at most $\sqrt{2}$.

Generalized Pigeonhole Principle
In the original pigeonhole principle, we think about having at least 2 pigeons in some hole. In the generalized version, we think about having at least n pigeons in a hole. Here is an example: we can show that in a class of 40 students, there are 3 students who were born in the same month. Suppose that is not true, then there are at most 2 students born in each month. The number of students is at most $12 \times 2 = 24$, which contradicts that there are 40 students. We may go further:

there are 4 students who share the same birth month, by the same reasoning: if there at most 3 students born in each month, there are at most $12 \times 3 = 36$ students in the class, which contradicts that there are 40 students. On the other hand, we may ask, what is the smallest number of students to guarantee that there are 5 students sharing the same birth month? The answer is $49 = 4(12) + 1$.

We state the **generalized pigeonhole principle** as follows:

If there are n pigeons and m pigeonholes, then there is a pigeonhole with at least $\lceil n/m \rceil$ pigeons.

Proof We prove by contradiction. Suppose there are fewer than $\lceil n/m \rceil$ pigeons in every pigeonhole. We look at two cases.

If n/m is an integer, then $\lceil n/m \rceil = n/m$, and the total number of pigeons is fewer than

$$m\left\lceil \frac{n}{m} \right\rceil = m\left(\frac{n}{m}\right) = n$$

This contradicts that there are n pigeons.

If n/m is not an integer, then having fewer than $\lceil n/m \rceil$ pigeons in every hole means that there are at most $\lfloor n/m \rfloor$ pigeons in every hole. Since n/m is not an integer, we have $\lfloor n/m \rfloor < n/m$, so again the total number of pigeons is fewer than

$$m\left(\frac{n}{m}\right) = n$$

contradicting that there are n pigeons.